# D1.2 Protein synthesis

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d1-2.html*

*Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 625–636 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## TRANSCRIPTION

### D1.2.12 Direction

pp. 625–626

RNA polymerase reads the template strand 3′ to 5′ and builds RNA **5′ to 3′**, adding nucleotides to the free 3′ -OH. Ribosomes read mRNA **5′ to 3′**, building the polypeptide from its amine end to its carboxyl end.

### D1.2.13 Starting at the promoter

pp. 626–627

General **transcription factors** bind the **promoter** just before a gene; RNA polymerase then joins them. Activators bound to enhancers can loop round and raise the rate. Without the right factors, the gene is not transcribed. RNA polymerase itself is not a transcription factor.

### D1.2.14 Non-coding DNA

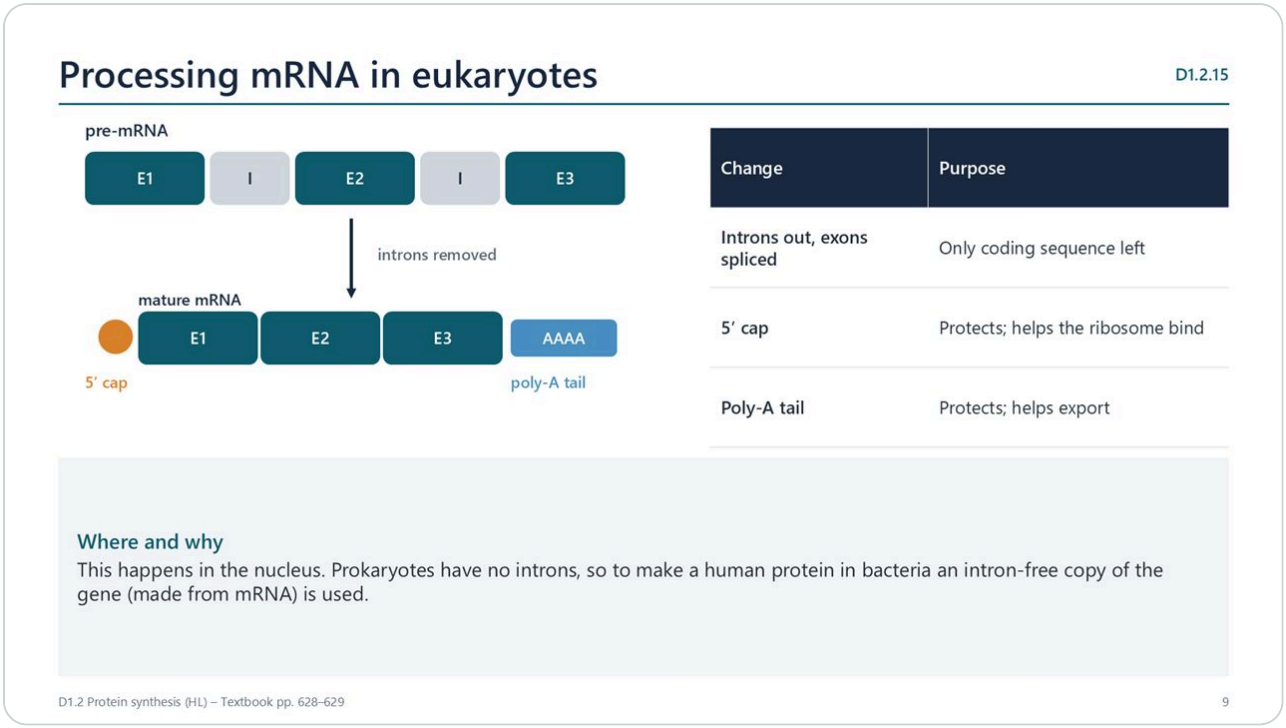
pp. 627–628

Only about 1.5% of human DNA codes for polypeptides. Non-coding sequences include regulators of gene expression (promoters, enhancers), **introns**, **telomeres** that protect chromosome ends during replication, and genes for **rRNA and tRNA**, whose RNA is the final product.

## PROCESSING

### D1.2.15 Post-transcriptional modification

pp. 628–629




In eukaryotes, **introns** are removed from pre-mRNA and **exons** spliced together; a **5′ cap** and a **3′ poly-A tail** are added, which protect the mRNA and help it leave the nucleus and bind ribosomes.

### D1.2.16 Alternative splicing

pp. 629–630

Different combinations of exons can be kept, so one gene can give several versions of a protein. The **troponin T** gene is spliced differently in foetal and adult heart muscle.

## TRANSLATION AND AFTER

### D1.2.17 Starting translation

pp. 630–633

The small subunit binds the 5′ end of mRNA and moves to the start codon AUG; the initiator tRNA (methionine, anticodon UAC) pairs with it; the large subunit joins, with the initiator tRNA in the **P site**. A tRNA with the next amino acid enters the **A site**, a peptide bond forms, the ribosome moves one codon, and the empty tRNA leaves from the **E site**.

### D1.2.18 Modifying polypeptides

pp. 633–635

Many polypeptides are modified before they work. **Pre-proinsulin** loses its signal sequence in the ER, giving proinsulin, which folds with three disulfide bonds; later the C-peptide is cut out, leaving **insulin**: A and B chains held by disulfide bonds.

### D1.2.19 Proteasomes

pp. 635–636

Damaged or unneeded proteins are tagged with **ubiquitin** and broken down by **proteasomes**. The amino acids are reused, so a working proteome depends on constant breakdown as well as synthesis.

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